

INSERTION SORT ALGORITHM

Array $\Rightarrow$ 5, 3, 4, 1, 2 [partially sorting array]

1st pass :- $i = 0 \rightarrow$ sort till index 1

5, 3, 4, 1, 2
↓

this is — 3, 5, 4, 1, 2
sorted

2nd pass :- $i = 1 \rightarrow$ sort till index 2

3, 5, 4, 1, 2
↓

this is — 3, 4, 5, 1, 2
sorted

3rd pass :- $i = 2 \rightarrow$ sort till index 3

3, 4, 5, 1, 2
↓

this is — 1, 3, 4, 5, 2
sorted

4th pass :- $i = 3 \rightarrow$ sort till index 4

1, 3, 4, 5, 2
↓

1, 2, 3, 4, 5 $\rightarrow$ sorted

Working 1 —

$i < n-2$ $j > 0$

$\begin{array}{cccccc} & & j & & & \\ 0 & 1 & 2 & 3 & 4 & \\ 5 & 3 & 4 & 1 & 2 & \end{array}$

$i = 0$

$j = 1$

$3 < 5 \Rightarrow \text{swap}$

3, 5, 4, 1, 2

→ sorted till index 1.

~~5, 3, 4~~, 3, 5, 4, 1, 2

$i = 1$

$j = 2$

$4 < 5 \Rightarrow \text{swap}$

3, 4, 5, 1, 2

→ sorted till index 2

$4 > 3 \Rightarrow \text{break}$

3, 4, 5, 1, 2

$i = 2$

$j = 3$

↓ $1 < 5 \Rightarrow \text{swap}$

3, 4, 1, 5, 2

↓ $1 < 4 \Rightarrow \text{swap}$

3, 1, 4, 5, 2

↓ $1 < 3 \Rightarrow \text{swap}$

1, 3, 4, 5, 2

→ sorted till index 3.

1, 3, 4, 5, 2

$i = 3$

$j = 4$

↓ $2 < 5 \Rightarrow \text{swap}$

1, 3, 4, 2, 5

↓ $2 < 4 \Rightarrow \text{swap}$

1, 3, 2, 4, 5

↓ $2 < 3 \Rightarrow \text{swap}$

1, 2, 3, 4, 5

↓ $2 > 1 \Rightarrow \text{Break}$

→ sorted till

index 4

Time Complexity :-

① Worst Case :- $O(n^2)$
[descending Order]

② Best Case :- $O(n)$
[already sorted]

• Why to use insertion sort?

* Adaptive :- steps get reduced if array is sorted.
i.e. no of swaps are reduced as compared to bubble sort.

* Stable Sorting Algorithm.

* Use for smaller values of n :- works good when array is partially sorted.